

Binary Polyhedral Groups in $SU(2)$

Paired lifts, parity cancellation, and direct matrix verification

Computational verification note

17 July 2026

Outcome. Explicit 2×2 complex matrix groups of orders 24, 48, and 120 were reconstructed for $2T$, $2O$, and $2I$. For each group, all 67 multivariate coefficients through total degree eight agree with the proposed paired-lift formula. Closure, unitarity, determinant one, the central element $-I$, and the odd-degree filter all pass.

1 The master coefficient formula

For a finite matrix group $G \leq GL(V)$,

$$\mathcal{S}_{G,V}(\mathbf{t}) = \frac{1}{|G|} \sum_{g \in G} \prod_{a \geq 1} \det(I - t_a g)^{-1}. \quad (1)$$

The identity

$$\det(I - tg)^{-1} = \sum_{d \geq 0} \chi_{\text{Sym}^d V}(g) t^d$$

shows that

$$[t_1^{\tau_1} \cdots t_\ell^{\tau_\ell}] \mathcal{S}_{G,V} = \dim \left(\bigotimes_a \text{Sym}^{\tau_a} V \right)^G. \quad (2)$$

Indeed, the coefficient is the trace of the Reynolds projector on the tensor product. Thus (1) and (2) are exact formal-power-series identities.

2 Proof of the binary-lift formula

Let $\tilde{\Gamma} \leq SU(2)$ be the inverse image of a finite rotation group $\Gamma \leq SO(3)$. If $R \in \Gamma$ rotates through angle θ , its two lifts q and $-q$ have spectra

$$(e^{i\theta/2}, e^{-i\theta/2}), \quad (-e^{i\theta/2}, -e^{-i\theta/2}).$$

Define

$$\begin{aligned} \Psi_\theta^+(\mathbf{t}) &= \prod_a (1 - 2 \cos(\theta/2) t_a + t_a^2)^{-1}, \\ \Psi_\theta^-(\mathbf{t}) &= \prod_a (1 + 2 \cos(\theta/2) t_a + t_a^2)^{-1}, \\ B_\theta &= \frac{1}{2}(\Psi_\theta^+ + \Psi_\theta^-). \end{aligned}$$

Pair the two terms in (1) over each $R \in \Gamma$. Because $|\tilde{\Gamma}| = 2|\Gamma|$, this gives

$$\mathcal{S}_{\tilde{\Gamma}, \mathbb{C}^2} = \frac{1}{|\Gamma|} \sum_{\theta} N_{\Gamma}(\theta) B_{\theta}. \quad (3)$$

This proves the proposed formula without requiring a separate conjugacy table for the binary group. Substituting the rotation counts yields

$$\mathcal{S}_{2T} = \frac{1}{12}(B_0 + 8B_{2\pi/3} + 3B_{\pi}), \quad (4)$$

$$\mathcal{S}_{2O} = \frac{1}{24}(B_0 + 6B_{\pi/2} + 8B_{2\pi/3} + 9B_{\pi}), \quad (5)$$

$$\mathcal{S}_{2I} = \frac{1}{60}(B_0 + 15B_{\pi} + 20B_{2\pi/3} + 12B_{2\pi/5} + 12B_{4\pi/5}). \quad (6)$$

3 The central parity filter

The matrix $-I$ belongs to each binary group. On $W_{\tau} = \bigotimes_a \text{Sym}^{\tau_a}(\mathbb{C}^2)$ it acts as $(-1)^{|\tau|}$. If $|\tau|$ is odd and v is invariant, then $v = (-I)v = -v$, so $v = 0$. Therefore

$$[m_{\tau}]\mathcal{S}_{2T} = [m_{\tau}]\mathcal{S}_{2O} = [m_{\tau}]\mathcal{S}_{2I} = 0 \quad \text{for odd } |\tau|. \quad (7)$$

The same cancellation is visible termwise in the average defining B_{θ} .

4 Independent matrix construction

The parent rotation matrices are constructed geometrically: signed permutation matrices give the octahedral group; the subgroup preserving the four vertices $(1, 1, 1), (1, -1, -1), (-1, 1, -1), (-1, -1, 1)$ gives the tetrahedral group; and compatible vertex-frame maps of the icosahedron give the icosahedral group.

For each rotation matrix, a standard trace-and-diagonal algorithm recovers one unit quaternion $q = w + xi + yj + zk$. It is mapped to

$$\rho(q) = \begin{pmatrix} w + iz & y + ix \\ -y + ix & w - iz \end{pmatrix} \in SU(2). \quad (8)$$

Both $\rho(q)$ and $-\rho(q)$ are retained. Since the sign choice for an individual lift is arbitrary but both signs are present, the resulting set is the full preimage and is independent of those choices.

This construction does not use (4)–(6). The implementation verifies all pairwise products against the reconstructed set, as well as unitarity, determinant one, and the presence of $-I$.

5 Verification protocol and results

For every partition τ through total degree eight, the direct route recursively computes complete homogeneous characters from all explicit matrices and averages their products. The independent route uses only the parent angle counts and paired half-angle spectra in (3). All odd rows are also checked separately against zero.

Finally, explicit matrices on $V \otimes V$ and $\text{Sym}^2(V)$ are averaged into Reynolds projectors. The alternating determinant tensor gives a one-dimensional invariant in $V \otimes V$, while $\text{Sym}^2(V)$ has no invariant vector.

Group	order	dimension	degree	checks	result
$2T$	24	2	0–8	67	PASS
$2O$	48	2	0–8	67	PASS
$2I$	120	2	0–8	67	PASS

All three sets are closed under multiplication. The largest observed unitarity error is 3.17×10^{-16} and the largest determinant error is 2.27×10^{-16} . Every odd-degree coefficient is exactly zero after integer validation.

For each group, the 4×4 Reynolds projector on $V \otimes V$ has rank and trace 1; its largest idempotence error is 1.04×10^{-16} . The symmetric-square projector has rank 0. Consequently the computation recovers the proposed low-degree distinction

$$[m_{(2)}] = 0, \quad [m_{(1,1)}] = 1.$$

6 Reproducibility and scope

Run from the repository root:

```
.venv/bin/python -m for_this_guy.matrix_group_formula_verification.binary_polyhedral_verification
```

Open `binary_polyhedral_verification.py` with marimo for interactive degree selection and complete row tables.

The finite checks validate the matrix lift, angle convention, class counts, central signs, and coefficient code in reasonable low dimensions. The all-degree conclusion is the formal proof (1)–(3); it does not depend on the chosen cutoff or floating-point experiment.